

Activity - 5

Date: 12/08/2026

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① Given input {4371, 1323, 6173, 4199, 4344, 9679, 1989} and a hash function $h(x) = x \pmod{10}$, show the

- Separate Chaining hash table
- open addressing hash table using linear probing
- open addressing hash table using quadratic probing
- open addressing hash table using second hash function $h_2(x) = 7 - (x \pmod{9})$.

a) Separate Chaining Hash Table.

$$h(x) = [x \% 10]$$

$$h(4371) = 4371 \% 10 = 1$$

$$h(1323) = 1323 \% 10 = 3$$

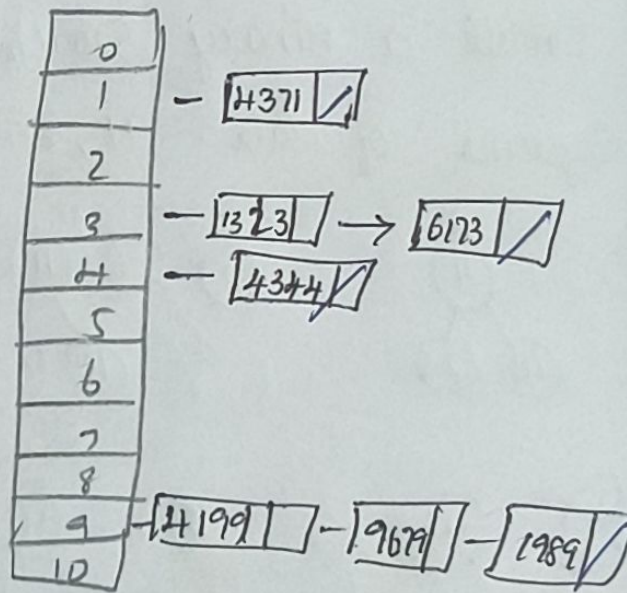
$$h(6173) = 6173 \% 10 = 3$$

$$h(4199) = 4199 \% 10 = 9$$

$$h(4344) = 4344 \% 10 = 4$$

$$h(9679) = 9679 \% 10 = 9$$

$$h(1989) = 1989 \% 10 = 9$$



b) open addressing hash Table using linear probing

$$h(\text{key}) = h(\text{key} + i) \text{ if } i \geq 0, 1, 2, 3,$$

$$i \geq 0, 1, 2, 3,$$

$$h(4371) = 4371 \% 10 = 1$$

$$h(1323) = 1323 \% 10 = 3$$

$$h(6173) = 6173 \% 10 = 3 \parallel$$

$$h(4199) = 4199 \% 10 = 9$$

$$h(4344) = 4344 \% 10 = 4 \parallel 5$$

$$h(9679) = 9679 \% 10 = 9 \parallel \rightarrow h(9679) = h(9679+1) = 9680$$

$$h(6173) = h(6173+1)$$

$$= 6174 \% 10 = 4 \checkmark$$

$$h(6173) = h(6173+2)$$

$$= 6175 \% 10 = 5$$

9679	0
4371	1
1992	2
1323	3
6174	4
4344	5
	6
	7
	8
4199	9

c) open addressing

$$h(\text{key}) = h(\text{key})$$

$$h(4371) = 1$$

$$h(1323) = 3$$

$$h(6173) = 3 \parallel$$

$$h(6173) = h(6173+1)$$

$$h(4199) = 9$$

$$h(4344) = 4 \parallel$$

$$h(4344) = h(4344+1)$$

$$h(9679) = 9$$

$$h(9679) = h(9679+1)$$

$$h(1989) = 9$$

$$h(1989) = h(1989+1)$$

$$1989+1$$

$$1989+1$$

d) open addressing

hash function.

$$h(\text{key}) = h(\text{key})$$

$$h(4371) = 1$$

$$h(1323) = 3$$

$$h(6173) = 3 \parallel$$

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199, 4344, 9679, 1989

mod 10, show the

hash table

using linear probing

using quadratic probing

using second hash.

Table

0	
1	4371
2	
3	1323 → 6173
4	4344
5	
6	
7	
8	
9	4199 → 9679 → 1989
10	

Table using linear probing

$i = 0, 1, 2, 3,$
 $h(6173) = h(6173 + 1)$
 $= 6174 \cdot 10 = 4 \checkmark$
 $h(6173) = h(6173 + 2)$
 $= 6175 \cdot 10 = 5$
 $h(9679) = h(9679 + 1) = 9680$

9679	0
4371	1
1992	2
1323	3
6174	4
4344	5
	6
	7
	8
4199	9

1989	0
4344	1
1992	2
1323	3
6174	4
4344	5
	6
	7
	8
4199	9

$$(4344 + 1) \cdot 10 = 4345 \cdot 10 = 43450$$

$$h(1989) = 1989 \cdot 10 = -9$$

$$1989 + 1 = 1990 = 0$$

$$1989 + 2 = 1991 = 1$$

$$1989 + 3 = 1992 = 2$$

c) open addressing hash Table using quadratic probe.

$$h(\text{Key}) = h(\text{Key} + i^2) \cdot 10$$

$$h(4371) = 1$$

$$h(1323) = 3$$

$$h(6173) = 3//$$

$$h(6173) = h(6173 + 1) \cdot 10 = 4 \checkmark$$

$$h(4199) = 9$$

$$h(4344) = 4//$$

$$h(4344) = h(4344 + 1) \cdot 10 = 5 \checkmark$$

$$h(9679) = 9$$

$$h(9679) = h(9679 + 1) \cdot 10 = 9680 = 0 \checkmark$$

$$h(1989) = 9$$

$$h(1989) = h(1989 + 1) \cdot 10 = 1990 = 0$$

$$1989 + 4 \cdot 10 = 1993 = 3$$

$$1989 + 9 \cdot 10 = 1998 = 8 \checkmark$$

9679	0
4371	1
1989	2
1323	3
6173	4
4344	5
	6
	7
1989	8
4199	9

d) open addressing hash Table using second hash function.

$$h(\text{key}) = h_1(\text{key}) + h_2(\text{key}) \cdot i$$

$$h(4371) = 1$$

$$h(1323) = 3$$

$$h(6173) = 3//$$

$$= 3 + (1 - 6173 \cdot 0.7) * 1$$

$$= 3 + 1 = 4$$

$$h(4199) = 9$$

$$h(4344) = 4 \parallel$$

$$= 4 + (1 - 4344 \cdot 0.7) * 1$$

$$= 4 + (1 - 4) * 1$$

$$= 4 + 3 * 1$$

$$= 7$$

$$h(9679) = 9 \parallel$$

$$= 9 + (1 - 9679 \cdot 0.7) * 1$$

$$= 9 + (1 - 5) * 1$$

$$= 9 + 2 * 1 = \underline{11}$$

$$= 9 + (1 - 5) * 2$$

$$= 9 + 2 * 2$$

$$= \underline{13}$$

$$h(1989) = 9 + (1 - 1989 \cdot 0.7) * 1$$

$$= 9 + 6 * 1$$

$$= 15 = \underline{51}$$

$$h(1989)$$

$$= 9 + (1 - 1989 \cdot 0.7) * 2$$

$$= 9 + 6 * 2$$

$$= 21 \cdot 10 = 1$$

$$= 9 + (1 - 1989 \cdot 0.7) * 3$$

	0
4371	1
	2
1323	3
6173	4
9679	5
	6
4344	7
	8
4199	9

$$= 9 + 6 * 3 = 27 \cdot 10 =$$

$$= 9 + 6 * 4 = 33 \cdot 10 =$$

$$= 9 + 6 * 6$$

Good
12/8

∴ Because the size of
excits.

	0
2531	1
	2
1323	3
6133	4
9639	5
	6
4544	7
4144	8
	9

$$= 9 + 6 * 3 = 27 + 10 = 7$$

$$= 9 + 6 * 4 = 33 + 10 = 3$$

$$= 9 + 6 * 6$$

Good
12/8

∴ Because the size of the 1989 doesn't
excite.

$$9 + (7 - 5) * 3$$

$$9 + 2 * 3$$

$$9 + 6 = 15$$